Ellis Franklin

PhD Student · Bioinformatics & Ecotoxicology

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EDUCATION

PhD — Bioinformatics & Ecotoxicology
Université de Lorraine
2023 — Present

Title: *"Analytical tool and toxicological interpretation keys for time- and dose-dependent transcriptomic responses."*
Supervisors: Prof. Marie Laure Delignette-Muller & Assoc. Prof. Elise Billoir.
Advisor: Assoc. Prof. Sophie Prud'homme.

Master — Biodiversity, Ecology, and Evolution
Université de Lyon 1
2021 — 2023

Topics: Biodiversity and functioning of ecosystems · Bioinformatics and Modelling in Ecology · Bayesian Statistics · Genomics in Ecology and Evolution · Parasitical and Mutualistic Interactions.

Bachelor — Life sciences
Université de Lyon 1
2018 — 2021

Topics: Community and Microbial Ecology · Ecophysiology · Genetics and Population Dynamics · Molecular tools for Ecology and Evolution.

RESEARCH EXPERIENCE

Doctoral Researcher
LIEC & LBBE
2023 — Present

Labs: LIEC & LBBE · **Institutions:** Université de Lorraine & VetAgroSup · **Project:** ANR Chroco
Ecotoxicogenomics thesis developing methodological approaches for analysis and interpretation of transcriptomic data modulated by contaminant dose and exposure time, across model (*Danio rerio*) and non-model (*Daphnia magna*) organisms.
Axis 1: cluefish — a workflow for structured biological interpretation of dose-response transcriptomic results (Franklin et al., 2025).
Axis 2: Extension of DRomics to time-dose-response data (in preparation, 2026).
More info: theses.fr/s370682.

Master's 2 Research Intern
LIEC, Université de Lyon 1
Jan — Jun 2023

Title: *"From modelling dose-response transcriptomic data to biological interpretation: an application to embryonic exposure of Danio rerio to DBP."*
Early development of cluefish, addressing redundancy limitations of functional enrichment by combining ORA with PPIN construction and clustering (STRING, Cytoscape), biological databases (KEGG, WikiPathways, AnimalTFDB), and dose-response modelling metrics (BMDs, curve trends).

Master's 1 Research Intern
LBBE, Université de Lyon 1
Apr — Jun 2022

Title: *"Development of a dose-response analysis on transcriptomic data to better understand the chronic effects of ionising radiation at low doses."*
First encounter with dose-response modelling applied to transcriptomic data, using DRomics to reanalyse RNA-seq data from *Danio rerio* embryos chronically exposed to ionising radiation. Addressed batch effect correction and explored the added value of a concentration-dependent framework, including BMD computation for ecological risk assessment.

PUBLICATIONS

Peer-reviewed articles

J. Ohanessian, S. M. Prud'homme, **E. Franklin**, G. Kitzinger, C. Lorgeoux, E. Billoir, V. Felten (2025). *Effects of dibutyl phthalate (DBP) on life history traits and population dynamics of Daphnia magna: comparison of two exposure regimes.* Peer Community Journal, 5. DOI · Preprint

E. Franklin, E. Billoir, P. Veber, J. Ohanessian, M. L. Delignette-Muller, S. M. Prud'homme (2025). *Cluefish: mining the dark matter of transcriptional data series with over-representation analysis enhanced by aggregated biological prior knowledge.* NAR Genomics and Bioinformatics, 7(3). DOI · Preprint · GitHub

COMMUNICATIONS

Oral — International

E. Franklin et al. (2024). *Interests and constraints towards unravelling endocrine disruptors mechanisms using dose-response transcriptomic data*. PRIMO22, Nantes, France. HAL

E. Franklin et al. (2025). *Exploring transcriptomic data with over-representation analysis enhanced by aggregated biological prior knowledge*. SETAC Europe, Vienna, Austria. HAL

E. Franklin et al. (2026). *Développement d'une méthode d'analyse intégrative de données transcriptomiques temps-dose-réponse* (ENG: *Development of an integrated analysis method for time-dose-repsonse transcriptomic data*). ECOBIM, Le Havre, France.

Oral — National

E. Franklin et al. (2023). *Chronologie des effets et des réactions face à une exposition à un contaminant*. GDR Ecotox Aqua 2023, Metz, France.

E. Franklin et al. (2024). *Cluefish: exploration of transcriptomic data with over-representation analysis enhanced by aggregated biological prior knowledge*. GDR Ecotox Aqua 2024, Grenoble, France.

E. Franklin et al. (2025). *Unravelling transcriptomic response dynamics in ecotoxicology*. GDR Ecostat, Metz, France.

Oral — Local & Internal

E. Franklin et al. (2026). *Cluefish: a workflow to make sense of transcriptomic dose-response data in ecotoxicology*. LBBE Internal Seminar, Lyon, France.

E. Franklin (2025). *Reproducible Research: Every little helps*. LEHNA Seminar, Lyon, France. Slides · GitHub

Posters

E. Franklin et al. (2024). *From dose-response modelling of transcriptomic data to biological interpretation: an application to embryonic exposure of zebrafish to DBP*. PRIMO22, Nantes, France. HAL

AWARDS

2024

Best Oral Presentation Prize, *PRIMO22 Scientific Committee (Ifremer)*, Nantes, France.

Best Illustration of Work, *DocDay, LIEC*, Nancy, France.

TEACHING & SUPERVISION

Teaching Assistant

Université de Lorraine

Practical work for Master 2 students (16h): **R programming, ggplot2, and Quarto.** *2024*

Workshop — LIEC

Metz & Nancy

A Step Towards Reproducible Research: Literate Programming. Full-day interactive workshop designed and delivered independently for LIEC lab members (ANTRE initiative, organised with Elise Billoir). Structured in two parts: a discussion-based introduction to reproducibility challenges, followed by a hands-on Quarto session. Workshop site · Materials *Feb 6, 2025*

Seminar — LBBE

Lyon

Reproducible Research: Every little helps? Half-day interactive seminar for the LBBE Student Seminar series (Les Pinsons Migrateurs). Covered the what, why, and how of reproducible research with discussion-driven format and interactive elements (Wooclap polls, drawing exercises). Seminar site · Materials *Feb 21, 2026*

Student Co-supervision

LIEC, Metz

Co-supervision of Master 2 internship: *"Chronic dose-response exposure of Daphnia magna to dibutyl phthalate at various exposure durations: development and application of a respirometry test and monitoring of eye development."* *2024*

TRAINING HIGHLIGHTS

Best practices for reproducible research in numerical ecology

CESAB-FRB & GDR EcoStat, Montpellier

Whole 38h training week co-organised by CESAB-FRB and GDR EcoStat. Covered the full reproducible research stack: project organisation, Git/GitHub, literate programming with Quarto, pipeline optimisation with `targets`, package development, dependency locking with `renv`, and containerisation with Docker. *Nov 20–24, 2023*

Reproducible research : Methodological Principles for Open Science

France Université Numérique / Inria

Success · 24h · Reproducible research principles, computational notebooks (Jupyter, RStudio), version control with GitLab. *2024*

SKILLS

Computational & Programming

Main languages: R (advanced), Python (beginner–intermediate), Bash
Reproducible research: Quarto, R Markdown, `renv`, `targets`, Git · GitHub / GitLab / Codeberg
Web & documents: HTML, CSS, Typst, Markdown
Software development: Developed cluefish — a structured, fully documented R workflow.

Bioinformatics

RNA-seq preprocessing: Salmon, Kallisto, HISAT2
Differential expression & dose-response: `DESeq2`, `edgeR`, `DRomics`
Annotation: biomaRt, KEGG, GO; identifier mapping across transcripts, genes, proteins
Functional enrichment (ORA, FCS, pathway topology): gprofiler2, clusterProfiler, topGO, DAVID; WGCNA
Network biology: PPIN construction and clustering (STRING, Cytoscape)
Data visualisation: `ggplot2`, `patchwork`, `plotly` integrated in Quarto reports.

Biological Interpretation

Functional knowledge: Fluent in KEGG, GO, WikiPathways, and related annotation sources; able to navigate and critically interpret enrichment outputs across biological levels (transcript ⟶ gene ⟶ protein ⟶ pathway ⟶ process)
Mechanistic reasoning: Experience deriving mechanistic hypotheses from omics perturbations — e.g. impaired retinol metabolism under dibutyl phthalate exposure in zebrafish embryos, followed by morphological validation (eye size) (Franklin et al., 2025) — bridging molecular signatures to experimentally testable organism-level outcomes.

Statistics & Modelling

Dose-response modelling: item selection (ANOVA, linear/quadratic trend tests, custom trend tests), curve fitting and model selection, BMD computation, non-parametric bootstrapping
Multivariate analyses, cluster analysis and trees.

Languages

English: Native · **French:** Fluent.

SERVICE

PhD Representative

LIEC, Université de Lorraine
2023 – 2025

Represented doctoral researchers at the lab level for two years.

Peer Review

Reviewer for *Nucleic Acids Research*.

Memberships

SFBI (Société Française de Bioinformatique) · GDR StatOmique · GDR EcoStat · SETAC Omics Interest Group

OUTREACH & SCIENCE COMMUNICATION

Workshop Facilitator

GDR Ecotox Aqua, France
2024

DRomics: Dose-Response Curves for Omics.

Fête de la Science

LIEC, Metz, France
Oct 10–12, 2024

Co-organised and facilitated a 3-day public outreach event titled *"Prédateurs et Proies : dans l'océan, l'exemple des requins et des maquereaux"*, aimed at school groups and general audiences. Designed and ran a board game and simulation game to explain predator–prey dynamics.